A bijective partition-theoretic interpretation for the Andrews-Merca truncated pentagonal number theorem

Xiangyu Ding and Lisa Hui Sun*

Center for Combinatorics, LPMC, Nankai University, Tianjin 300071, China
dingmath@mail.nankai.edu.cn sunhui@nankai.edu.cn

Abstract

Since Andrews and Merca established a truncated version of Euler's pentagonal number theorem in 2012, truncated theta series have developed into an active subfield within q-series and partition theory. Truncated variants of many classical theta identities, together with the non–negativity of their q-coefficients, partition–theoretic interpretations, and related partition inequalities have been extensively studied. In this paper, for Andrews and Merca's truncated pentagonal number theorem, we further give an alternative partition–theoretic interpretation and construct a weight–preserving bijection onto the partitions obtained by Andrews and Merca. Furthermore, by reparameterizing Rowell's conjugate Bailey pair, we obtain a conjugate WP–Bailey pair along with a transformation formula. This formula specializes to the Andrews–Merca truncated pentagonal-number identity, as well as the companion formulas for the truncated Gauss square-number and triangular-number identities.

Keywords. truncated theta series; partitions; Gaussian polynomials; bijections; WP–Bailey pairs

2020 Mathematics Subject Classification. 05A17, 11P81, 33D15

1 Introduction

Euler's pentagonal number theorem

$$\sum_{j=-\infty}^{\infty}(-1)^j q^{j(3j+1)/2}=(q;q)_\infty \tag{1.1}$$

is a classical bridge between the theta series and the integer partitions. Multiplying it by the generating function for the partition functions

$$\sum_{n\geq 0}p(n)q^n=\frac{1}{(q;q)_\infty}$$

yields the well-known recurrence relation to compute the partition function $p(n)$, see, for example, [2]. Another two closely related formulas are Gauss square-number and triangular-number theta identities

$$\sum_{j=-\infty}^{\infty}(-1)^j q^{j^2}=\frac{(q;q)_\infty}{(-q;q)_\infty} \tag{1.2}$$

*Corresponding author

and

$$\sum_{j=-\infty}^{\infty}(-1)^j q^{2j^2+j}=\frac{(q^2;q^2)_\infty}{(-q;q^2)_\infty}. \tag{1.3}$$

These identities (1.1)–(1.3) are immediate consequences of the Jacobi triple product identity [13, Eq. (II.28)]. Their finite truncations exhibit positivity phenomena that do not appear in the original infinite–sum identities. When a finite segment of a theta series is truncated and normalized by the corresponding product side, the resulting coefficients commonly possess a uniform sign. This naturally motivates the problem of deriving explicit nonnegative expansions for the normalized truncated sums and, whenever feasible, of providing combinatorial explanations via partitions or direct bijections.

In 1951, Shanks [19] obtained a truncated version of Euler's pentagonal number theorem, while it didn't address the positivity of the coefficients. Andrews and Merca [4, Lemma 1.2] revisited Shanks' truncated sum and obtained the following expression

$$\frac{1}{(q;q)_\infty}\sum_{j=-k}^{k-1}(-1)^j q^{j(3j+1)/2}=1+(-1)^{k-1}\sum_{n\geq k}\frac{q^{\binom{k}{2}+(k+1)n}}{(q;q)_n}\begin{bmatrix}n-1\\k-1\end{bmatrix}, \tag{1.4}$$

where $k\geq 1$. The coefficient of q^n in the sum on the right hand side can be interpreted as the number of partitions in $\mathcal{M}_k(n)$ [4, Theorem 1.1], which is the set of partitions of n with k being the least missing positive part and the number of parts larger than k exceeding the number of parts smaller than k. Thereby

$$\sum_{n\geq 0}|\mathcal{M}_k(n)|q^n=\sum_{n\geq k}\frac{q^{\binom{k}{2}+(k+1)n}}{(q;q)_n}\begin{bmatrix}n-1\\k-1\end{bmatrix}=:B_k(q). \tag{1.5}$$

Later, Kolitsch and Burnette [16] provided another interpretation of the coefficients in terms of partition pairs.

Guo and Zeng [14] gave the following trucated versions for Gauss square-number and triangular-number theta identities (1.2) and (1.3) as follows

$$\begin{aligned}&\frac{(-q;q)_\infty}{(q;q)_\infty}\sum_{j=-k}^{k}(-1)^j q^{j^2}\\&\quad=1+(-1)^k\sum_{n\geq k+1}\frac{(-q;q)_k(-1;q)_{n-k}q^{(k+1)n}}{(q;q)_n}\begin{bmatrix}n-1\\k\end{bmatrix}.\\&\frac{(-q;q^2)_\infty}{(q^2;q^2)_\infty}\sum_{j=0}^{k-1}(-1)^j(1-q^{2j+1})q^{j(2j+1)}\\&\quad=1+(-1)^{k-1}\sum_{n\geq k}\frac{(-q;q^2)_k(-q;q^2)_{n-k}q^{2(k+1)n-k}}{(q^2;q^2)_n}\begin{bmatrix}n-1\\k-1\end{bmatrix}_{q^2}.\end{aligned}$$

Andrews and Merca subsequently used the Rogers–Fine identity to treat the Euler's and Gauss's truncations in [5]. One of their square-number formulas is

$$\frac{(-q;q)_\infty}{(q;q)_\infty}\sum_{j=-k}^{k}(-1)^j q^{j^2}$$

$$= 1 + 2(-1)^k \frac{(-q;q)_k}{(q;q)_k} \sum_{r\geq 0} \frac{q^{(k+1)(k+r+1)}(-q^{k+r+2};q)_\infty}{(1-q^{k+r+1})(q^{k+r+2};q)_\infty}. \tag{1.7}$$

Further refinements and companion identities were obtained by Yee, Mao, Wang–Yee, and Xia–Yee–Zhao, et al., see [17,21,23,24].

On the other side, Bailey pairs and their well-poised analogues provide a systematic mechanism for transforming the tails of theta series, see, for example [1,3,7]. Ding and Sun [9] applied the Bailey lattice of Agarwal, Andrews, and Bressoud [1] to obtain an Andrews–Gordon type family of truncated Jacobi triple product series, which include the following truncated Euler's pentagonal number series:

$$\frac{1}{(q;q)_\infty} \sum_{j=-k}^{k-1} (-1)^j q^{j(3j+1)/2}$$
$$= 1 + (-1)^{k-1} \sum_{r\geq 0} \frac{q^{(2k+1)r+r^2+(3k^2+k)/2}(1-q^k)}{(q;q)_r (q;q)_{r+2k}(1-q^{r+k})}. \tag{1.8}$$

The truncated sum in (1.8) is in the same form as the Andrews–Merca formula (1.4), but its expansion form is different, for which we denote

$$A_k(q) = \sum_{r\geq 0} \frac{q^{(2k+1)r+r^2+(3k^2+k)/2}(1-q^k)}{(q;q)_r (q;q)_{r+2k}(1-q^{r+k})} = \sum_{m\geq k} \frac{q^{m(m+1)+\binom{k}{2}}}{(q;q)_m (q^{k+1};q)_m} \begin{bmatrix} m-1 \\ k-1 \end{bmatrix}, \tag{1.9}$$

where the second sum is obtained by substituting $r = m - k$.

Let $\mathcal{R}(r,b)$ denote the set of partitions fitting in an $r \times b$ rectangle, that is,

$$\mathcal{R}(r,b) = \{\lambda = (\lambda_1, \ldots, \lambda_r) : b \geq \lambda_1 \geq \cdots \geq \lambda_r \geq 0\},$$

where $r, b > 0$. Let $\ell(\lambda)$ denote the number of positive parts of λ. Denote $\mathcal{P}_{\leq m}$ and $\mathcal{P}_{[a,b]}$ the sets of partitions whose parts are at most m and lie in $[a,b]$, respectively. Define a set of partition triples as follows

$$\mathcal{A}_k(N) = \big\{(\alpha,\eta,\rho)\colon \alpha \in \mathcal{P}_{\leq m},\ \eta \in \mathcal{P}_{[k+1,k+m]},\ \rho \in \mathcal{R}(k-1, m-k),$$
$$\text{with } m \geq k \text{ satisfying that } \ m(m+1) + \tbinom{k}{2} + |\alpha| + |\eta| + |\rho| = N\big\}.$$

Note that for given k, N and the partition triple (α,η,ρ), the integer m in the above set is uniquely determined by

$$m(m+1) = N - \binom{k}{2} - |\alpha| - |\eta| - |\rho|. \tag{1.10}$$

It is known from [20, Chapter 1] that the classical partition interpretation of the Gaussian polynomial is

$$\begin{bmatrix} n \\ r \end{bmatrix} = \sum_{\lambda \in \mathcal{R}(r,n-r)} q^{|\lambda|}. \tag{1.11}$$

Therefore, it follows from (1.9) that

$$A_k(q) = \sum_{N\geq 0} |\mathcal{A}_k(N)|q^N.$$

Analytically, it direct to see that $A_k(q) = B_k(q)$ by comparing (1.4) and (1.8). Combinatorially, our main result is to construct a bijection between the set of partition triples $\mathcal{A}_k(N)$ and the Andrews–Merca partition set $\mathcal{M}_k(N)$.

Theorem 1.1. *For every $k \geq 1$ and $N \geq 0$, there is a weight-preserving bijection*

$$\begin{aligned}\omega : \mathcal{A}_k(N) &\longrightarrow \mathcal{M}_k(N)\\ (\alpha, \eta, \rho) &\longmapsto \pi\end{aligned}$$

and for m being uniquely determined by (1.10), it holds

$$m(m+1) + \binom{k}{2} + |\alpha| + |\eta| + |\rho| = |\pi| = N.$$

The key ingredient of this bijection is as follows. Once m is determined, the Gaussian refactorization algorithm in Lemma 3.1 is applied to transform partition pair (η, ρ) to obtain (σ, τ). Using the k-Durfee rectangle, we then decompose and recombine the resulting partition triple to obtain an Andrews–Merca partition by using the staircase operations.

Furthermore, from Rowell's general conjugate Bailey pair [18, Theorem 2.1], we construct a conjugate WP–Bailey pair and derive a transformation formula. One specialization recovers (1.8), which gives an alternative WP–Bailey derivation of the truncated Euler's pentagonal number theorem. Two further specializations yield companion expressions for the truncated Gauss square-number and triangular-number series.

The rest of this paper is organized as follows. Section 2 introduces some basic definitions and notation. Section 3 explains the Gaussian refactorization algorithm which gives a partition theoretical interpretation of a transformation identity for products of two Gaussian polynomials. Section 4 combines Gaussian refactorization algorithm with the k-Durfee rectangle to construct the bijection ω. Section 5 develops the conjugate WP–Bailey transformation and its Euler and Gauss type specializations.

2 Preliminaries

Throughout this paper, we assume that $0 < |q| < 1$ and the parameters satisfy that the corresponding denominators do not vanish and the infinite sums are absolutely convergent. The standard notations and identities used below may be found in [13].

The q-shifted factorial is defined by

$$(a;q)_n = \prod_{n=0}^{n-1}(1-aq^n),$$

for $n \geq 1$ and $(a;q)_0 = 1$. We also use the notation $(a;q)_\infty = \prod_{n=0}^{\infty}(1-aq^n)$, and $(a_1, \ldots, a_m; q)_n = \prod_{i=1}^{m}(a_i;q)_n$. The Gaussian polynomial, or the q-binomial coefficient is

given by

$$\begin{bmatrix} n \\ r \end{bmatrix} = \begin{cases} \dfrac{(q;q)_n}{(q;q)_r(q;q)_{n-r}}, & 0 \le r \le n, \\ 0, & \text{otherwise.} \end{cases}$$

Except the partition interpretation (1.11), the Gaussian polynomial also admits a geometric interpretation via the Grassmannian $\mathrm{Gr}(r, V)$:

$$\begin{bmatrix} n \\ r \end{bmatrix} = |\mathrm{Gr}(r, V)(\mathbb{F}_q)|,$$

where $\dim_{\mathbb{F}_q} V = n$. See [12, Chapter 9] or [15] for its Schubert-cell decomposition.

Recall that a partition of a nonnegative integer n is a finite nonincreasing sequence of positive integers $\lambda_1, \lambda_2, \ldots, \lambda_k$ such that $\sum_{i=1}^{k} \lambda_i = n$, see, for example, [2]. The Young diagram of λ is a left-justified array of unit squares, where the i-th row contains exactly λ_i unit squares. We write λ' for the conjugate partition of λ, where $\lambda' = (\lambda'_1, \lambda'_2, \ldots,)$ is defined by

$$\lambda'_i := \#\{j \ge 1 \mid \lambda_j \ge i\},$$

for each integer $i \ge 1$. Geometrically, λ' is obtained by transposing the Young diagram of λ, interchanging rows and columns.

Following Chu [8] (see also [11]), for a fixed nonnegative integer k, the k-Durfee rectangle of a partition λ is the largest rectangle of the form $h \times (h + k)$ contained in its Young diagram. Equivalently,

$$h = \max\{r \ge 1 : \lambda_r \ge r + k\}.$$

By convention, we set $h = 0$ if the above set is empty. Writing $m = h + k$, the rectangle has size

$$(m - k) \times m.$$

If $m > k$, then $\lambda_{m-k} \ge m$, whereas the maximality always gives $\lambda_{m-k+1} \le m$. For $k = 0$ this reduces to the ordinary Durfee square. See Figure 1 for the Young diagram of a partition and its 3-Durfee rectangle.

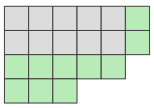


Figure 1: The 3-Durfee rectangle of size 2×5.

3 The Gaussian refactorization algorithm

In this section, we provide the Gaussian refactorization algorithm which transforms partition pairs restricted to two rectangles into partition pairs subject to two other rectangular constraints. Consequently, it gives a bijective proof of the following Gaussian refactorization

$$\begin{bmatrix} m+s-1 \\ s \end{bmatrix} \begin{bmatrix} m-1 \\ k-1 \end{bmatrix} = \begin{bmatrix} m+s-1 \\ k-1 \end{bmatrix} \begin{bmatrix} m+s-k \\ s \end{bmatrix}. \tag{3.1}$$

It can also be proved by counting the $\mathbb{F}_q$-rational points of the partial flag variety

$$\mathrm{Fl}(k-1, m-1; V) = \{\, U \subset W \subset V : \dim U = k-1,\ \dim W = m-1 \,\},$$

where $\dim_{\mathbb{F}_q} V = m+s-1$. Counting first over W and then by $U \subset W$ gives

$$\begin{bmatrix} m+s-1 \\ s \end{bmatrix} \begin{bmatrix} m-1 \\ k-1 \end{bmatrix},$$

whereas counting first over U and then by $W \setminus U$ gives

$$\begin{bmatrix} m+s-1 \\ k-1 \end{bmatrix} \begin{bmatrix} m+s-k \\ s \end{bmatrix}.$$

Thus the two sides of (3.1) enumerate the same partial flags in different orders.

Let $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_r) \in \mathcal{R}(r, b)$ in which the zero parts are allowed. For $r > 0$, denote the staircase shift operator to the partition λ by

$$\beta^+(\lambda) = \{\lambda_r, \lambda_{r-1}+1, \ldots, \lambda_1 + r - 1\},$$

which is an r-element subset of $\{0, 1, \ldots, r+b-1\}$ and

$$|\lambda| = \sum_{x \in \beta^+(\lambda)} x - \binom{r}{2}. \tag{3.2}$$

For the complement of λ in the $r \times b$ rectangle, we also set

$$\beta^-(\lambda) = \{b - \lambda_1,\ b - \lambda_2 + 1, \ldots, b - \lambda_r + r - 1\}.$$

It is again an r-element subset of $\{0, 1, \ldots, r+b-1\}$ and

$$|\lambda| = rb + \binom{r}{2} - \sum_{x \in \beta^-(\lambda)} x. \tag{3.3}$$

When $r = 0$, we regard both $\beta^+(\lambda)$ and $\beta^-(\lambda)$ as empty. By using the above two staircase shift operators, we derive a bijection between the sets of the partition pairs corresponding to both sides of (3.1).

Lemma 3.1. *For $m \geq k \geq 1$ and $s \geq 0$, there is a weight-preserving bijection*

$$\begin{aligned} \psi : \mathcal{R}(s, m-1) \times \mathcal{R}(k-1, m-k) &\longrightarrow \mathcal{R}(k-1, m+s-k) \times \mathcal{R}(s, m-k) \\ (\mu, \rho) &\longmapsto (\sigma, \tau). \end{aligned}$$

Proof. Let

$$U = \{0, 1, \ldots, m+s-2\}, \qquad A = \beta^-(\mu), \qquad B = \beta^+(\rho).$$

The set $\{0, 1, \ldots, t\}$ is treated to be empty when $t < 0$. Thus $A \subseteq U$ has cardinality s, while $B \subseteq \{0, 1, \ldots, m-2\}$ has cardinality $k-1$. Write

$$U \setminus A = \{a_0 < a_1 < \cdots < a_{m-2}\}.$$

Select the elements indexed by B in this ordered complement and denote

$$C = \{a_b : \ b \in B\}.$$

The set C has $k-1$ elements and therefore determines a unique partition $\sigma \in \mathcal{R}(k-1, m+s-k)$ such that $\beta^+(\sigma) = C$.

Next write

$$U \setminus C = \{c_0 < c_1 < \cdots < c_{m+s-k-1}\}.$$

Since $C \subseteq U \setminus A$, the sets A and C are disjoint. The elements of A therefore occur in the ordered complement $U \setminus C$. Record their positions by

$$D = \{i : \ c_i \in A\}.$$

The set D has s elements and therefore determines a unique $\tau \in \mathcal{R}(s, m-k)$ satisfying $\beta^-(\tau) = D$. Define

$$\psi(\mu, \rho) = (\sigma, \tau).$$

Noting that every step of the above procedure is obviously invertible, hence ψ is a bijection.

To verify the weight preservation, write $\Sigma(X) = \sum_{x \in X} x$ for brevity. Since C is selected from the ordered set $U \setminus A$, it is easy to see that

$$\Sigma(C) = \Sigma(B) + I, \qquad I = \#\{(a, c) \in A \times C : \ a < c\}.$$

Likewise, D records the positions of the elements of A after those of C have been deleted, and hence

$$\Sigma(D) = \Sigma(A) - J, \qquad J = \#\{(a, c) \in A \times C : \ c < a\}.$$

Since $A \cap C = \varnothing$, every pair $(a, c) \in A \times C$ contributes to exactly once to I or J, and thereby $I + J = s(k-1)$. Therefore

$$\Sigma(C) - \Sigma(D) = \Sigma(B) - \Sigma(A) + s(k-1). \tag{3.4}$$

Using (3.2) and (3.3), we obtain

$$|\mu| = s(m-1) + \binom{s}{2} - \Sigma(A), \qquad |\rho| = \Sigma(B) - \binom{k-1}{2},$$

$$|\sigma| = \Sigma(C) - \binom{k-1}{2}, \qquad |\tau| = s(m-k) + \binom{s}{2} - \Sigma(D).$$

Together with (3.4), these identities give $|\mu| + |\rho| = |\sigma| + |\tau|$ and thereby ψ is weight-preserving. □

Example 3.2. Take $k = 3$, $m = 5$, $s = 2$, $\mu = (4, 1)$, and $\rho = (2, 1)$. Then $(\mu, \rho) \in \mathcal{R}(2, 4) \times \mathcal{R}(2, 2)$,

$$A = \beta^-(\mu) = \{0, 4\}, \qquad B = \beta^+(\rho) = \{1, 3\}, \qquad U = \{0, 1, 2, 3, 4, 5\}.$$

Since $U \setminus A = \{1, 2, 3, 5\}$, selecting the elements indexed by $B = \{1, 3\}$ gives $C = \{2, 5\}$. Then from $\beta^+(\sigma) = \{2, 5\}$, it follows that $\sigma = (4, 2)$. In the ordered complement $U \setminus C = \{0, 1, 3, 4\}$,

the elements of $A = \{0, 4\}$ occupy the positions $D = \{0, 3\}$. Hence $\beta^-(\tau) = \{0, 3\}$ and $\tau = (2)$, so

$$\psi((4,1),(2,1)) = ((4,2),(2)).$$

It is easy to check that $|\mu| + |\rho| = |\sigma| + |\tau| = 8$.

It is natural to ask whether the Gaussian refactorization algorithm stated in this section can be extended to the case of q-multinomial coefficients, which could serve as a starting point for bijective models of Andrews–Gordon-type identities and their Bailey-lattice truncations appeared in the related works such as [1, 6, 9, 10, 22].

4 The bijection between $\mathcal{A}_k(N)$ and $\mathcal{M}_k(N)$

Based on the Gaussian refactorization algorithm, we are ready to construct the bijection between the set of partition triples $\mathcal{A}_k(N)$ and the Andrews–Merca partition set $\mathcal{M}_k(N)$.

Proof of Theorem 1.1. Taking $(\alpha, \eta, \rho) \in \mathcal{A}_k(N)$, then the integer $m \geq k$ can be determined uniquely from (1.10). We now construct $\omega(\alpha, \eta, \rho)$ as follows.

Step 1: The Gaussian refactorization algorithm. Let $s = \ell(\eta)$ and $n = m + s$. Since every part of η lies in $[k + 1, k + m]$, write

$$\eta_i = k + 1 + \mu_i \qquad (1 \leq i \leq s),$$

where $\mu \in \mathcal{R}(s, m - 1)$ and $|\eta| = s(k + 1) + |\mu|$. Apply the bijection given in Lemma 3.1 to obtain

$$(\sigma, \tau) = \psi(\mu, \rho) \in \mathcal{R}(k - 1, n - k) \times \mathcal{R}(s, m - k),$$

with

$$|\mu| + |\rho| = |\sigma| + |\tau|.$$

Step 2: Rearrangement into a partition. The partition σ remains unchanged. Starting with an $(m - k) \times m$ rectangle, attach τ' to its right and place α below it. Denote the resulting partition by Λ, and set $\lambda = \Lambda'$. Since $\tau' \in \mathcal{R}(m - k, s)$, the largest part of Λ is at most $m + s = n$. Hence $\ell(\lambda) \leq n$. Moreover,

$$|\lambda| = |\Lambda| = m(m - k) + |\tau| + |\alpha|.$$

It is obvious that $(m-k) \times m$ is the k-Durfee square of Λ. We can see that n, λ, and σ therefore satisfy that

$$\begin{aligned}
\binom{k}{2} + (k+1)n + |\lambda| + |\sigma| &= \binom{k}{2} + (k+1)(m+s) + m(m-k) + |\alpha| + |\tau| + |\sigma| \\
&= \binom{k}{2} + m(m+1) + s(k+1) + |\alpha| + |\mu| + |\rho| \\
&= \binom{k}{2} + m(m+1) + |\alpha| + |\eta| + |\rho| = N. \qquad (4.1)
\end{aligned}$$

Step 3: Recombination of the Andrews–Merca partition. Pad λ with zeros to length n and define

$$\omega(\alpha,\eta,\rho) = (\lambda_1 + k + 1, \ldots, \lambda_n + k + 1) \sqcup (k-1, k-2, \ldots, 1) \sqcup \sigma', \tag{4.2}$$

where $\sqcup$ denotes multiset union of the elements followed by rearranging in weakly decreasing order. For $k = 1$, the staircase $(k-1, k-2, \ldots, 1)$ is empty. The first block in (4.2) consists of exactly n parts larger than k, whereas the other two blocks contain only parts smaller than k. The second block contributes one copy of each integer $1, 2, \ldots, k-1$. Furthermore, $\sigma \in \mathcal{R}(k-1, n-k)$ implies that σ' has at most $n-k$ parts and each part is at most $k-1$. Thus k is the least missing part and the number of parts smaller than k is at most

$$(k-1) + (n-k) = n - 1.$$

Consequently, $\omega(\alpha,\eta,\rho) \in \mathcal{M}_k(N)$, and (4.1) gives

$$|\omega(\alpha,\eta,\rho)| = |\lambda| + (k+1)n + \binom{k}{2} + |\sigma| = N.$$

Noting that each of the above steps is invertible, so ω is a bijection and weight-preserving. □

We use the following example to illustrate the detailed steps of the bijection ω.

Example 4.1. Let $k = 3$, $N = 58$, and

$$(\alpha,\eta,\rho) = ((5,3,1),(8,5),(2,1)) \in \mathcal{A}_3(58).$$

The auxiliary parameter $m = 5$ which is derived from

$$m(m+1) = 58 - \binom{3}{2} - |(5,3,1)| - |(8,5)| - |(2,1)| = 30.$$

Step 1. Since $k+1 = 4$, write $\eta = (4+4, 4+1)$, so $\mu = (4,1)$, $s = \ell(\eta) = 2$, and $n = m+s = 7$. The Gaussian refactorization algorithm as shown in Example 3.2 gives

$$\psi: \ (\mu,\rho) = ((4,1),(2,1)) \longmapsto (\sigma,\tau) = ((4,2),(2)).$$

Step 2. Take the 3-Durfee rectangle of size $(m-k) \times m = 2 \times 5$. Attaching $\tau' = (1,1)$ to its right and placing $\alpha = (5,3,1)$ below it gives

$$\Lambda = (6,6,5,3,1), \qquad \lambda = \Lambda' = (5,4,4,3,3,2).$$

Thus the relation (4.1) holds, that is,

$$\binom{3}{2} + 4 \cdot 7 + |\lambda| + |\sigma| = 3 + 28 + 21 + 6 = 58.$$

Step 3. Padding λ with one zero to make it of length 7, adding $k+1 = 4$ to all these seven entries, and inserting the staircase $(2,1)$ together with $\sigma' = (4,2)' = (2,2,1,1)$, it gives

$$\pi = (\lambda_1 + 4, \ldots, \lambda_7 + 4) \sqcup (2,1) \sqcup (4,2)'$$

$$= (9, 8, 8, 7, 7, 6, 4, 2, 2, 2, 1, 1, 1).$$

Here 3 is the least missing part, and π has seven parts larger than 3 and six parts smaller than 3. Hence $\pi \in \mathcal{M}_3(58)$.

Figure 2 shows the detailed steps of the bijection $\omega(\alpha, \eta, \rho) = \pi$. It also can be checked that during the main steps the weights satisfy

$$m(m+1) + \binom{k}{2} + |\alpha| + |\eta| + |\rho| = (k+1)n + \binom{k}{2} + |\Lambda| + |\sigma| = |\pi| = N.$$

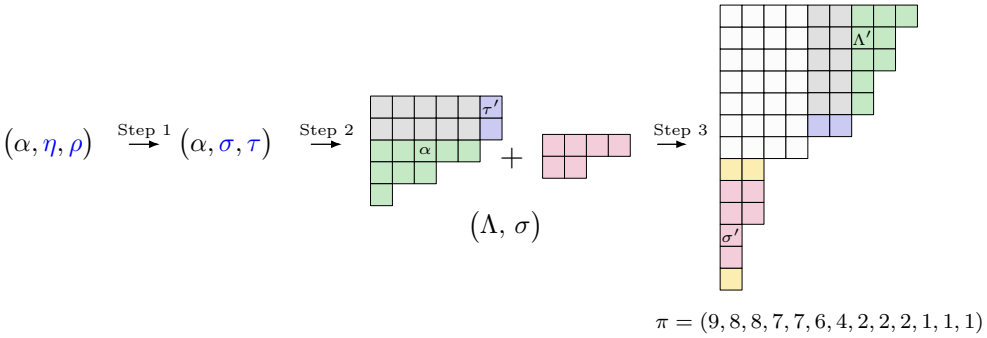


Figure 2: The steps of the bijection in Example 4.1.

5 A WP–Bailey transformation

In this section, by constructing a WP–Bailey transformation, we develop the analytic counterpart of the bijection ω between $\mathcal{A}_k(N)$ and $\mathcal{M}_k(N)$.

Recall that a pair of sequences $(\alpha_n(a), \beta_n(a))_{n\geq 0}$ is a Bailey pair relative to a if

$$\beta_n(a) = \sum_{r=0}^{n} \frac{\alpha_r(a)}{(q;q)_{n-r}(aq;q)_{n+r}}. \tag{5.1}$$

A conjugate Bailey pair relative to a is a pair $(\gamma_n(a), \delta_n(a))_{n\geq 0}$ satisfying

$$\gamma_n(a) = \sum_{r=n}^{\infty} \frac{\delta_r(a)}{(q;q)_{r-n}(aq;q)_{r+n}}. \tag{5.2}$$

For these basic definition, see [7, 18].

The well-poised analogue of Bailey pairs needs an additional parameter, we use the symbol K, to avoid confusion with the integer k used above. A WP–Bailey pair relative to (K, a) is a pair $(\alpha_n(K, a), \beta_n(K, a))_{n\geq 0}$ such that

$$\beta_n(K, a) = \sum_{r=0}^{n} \frac{(K/a;q)_{n-r}(K;q)_{n+r}}{(q;q)_{n-r}(aq;q)_{n+r}} \alpha_r(K, a). \tag{5.3}$$

A conjugate WP–Bailey pair relative to (K,a) is defined similarly by

$$\gamma_n(K,a)=\sum_{r=n}^{\infty}\frac{(K/a;q)_{r-n}(K;q)_{r+n}}{(q;q)_{r-n}(aq;q)_{r+n}}\delta_r(K,a). \tag{5.4}$$

When $K=0$, (5.3) reduces to the case of the ordinary Bailey pairs (5.1). Substituting (5.4) into the left-hand side of (5.5) and interchanging the order of the summations gives the following WP–Bailey lemma, see also [3].

Lemma 5.1 (WP–Bailey lemma)**.** *Let $(\alpha_n,\beta_n)_{n\geq 0}$ be a WP–Bailey pair and $(\gamma_n,\delta_n)_{n\geq 0}$ be a conjugate WP–Bailey pair, both relative to (K,a). Then*

$$\sum_{n\geq 0}\alpha_n(K,a)\gamma_n(K,a)=\sum_{n\geq 0}\beta_n(K,a)\delta_n(K,a). \tag{5.5}$$

We shall start from the following Rowell's general conjugate Bailey pair (Γ_n,Δ_n) relative to a [18, Theorem 2.1]:

$$\begin{aligned}\Gamma_n&=\frac{(Kq/b,Kq/c;q)_\infty}{(Kq/bc,K;q)_\infty}\frac{(d,K;q)_n}{(aq/d,aq;q)_n}\left(-\frac{1}{d}\right)^n q^{-\binom{n}{2}}\\&\quad\times\sum_{r=n}^{\infty}\frac{(K/a;q)_{r-n}(Kq^n;q)_r}{(q;q)_{r-n}(aq^{n+1};q)_r}\frac{(aq/d,b,c;q)_r}{(Kq/b,Kq/c,Kd/a;q)_r}(1-Kq^{2r})\left(-\frac{Kd}{bcq}\right)^r q^{r(r+3)/2},\\ \Delta_n&=\frac{(d,b,c;q)_n}{(Kd/a;q)_n}\left(\frac{Kq}{bc}\right)^n.\end{aligned} \tag{5.6}$$

The following theorem lifts Rowell's conjugate Bailey pair to a conjugate WP–Bailey pair.

Theorem 5.2. *Provided that*

$$\delta_n(K,a)=\frac{(aq/d,b,c;q)_n}{(Kq/b,Kq/c,Kd/a;q)_n}(1-Kq^{2n})\left(-\frac{Kd}{bcq}\right)^n q^{n(n+3)/2}, \tag{5.7}$$

$$\begin{aligned}\gamma_n(K,a)&=\frac{(Kq/bc,K;q)_\infty}{(Kq/b,Kq/c;q)_\infty}\frac{(aq/d;q)_n}{(d;q)_n}(-d)^n q^{\binom{n}{2}}\\&\quad\times\sum_{r=n}^{\infty}\frac{1}{(q;q)_{r-n}(aq;q)_{r+n}}\frac{(d,b,c;q)_r}{(Kd/a;q)_r}\left(\frac{Kq}{bc}\right)^r,\end{aligned} \tag{5.8}$$

then $(\gamma_n(K,a),\delta_n(K,a))$ is a conjugate WP–Bailey pair relative to (K,a). Consequently, for every WP–Bailey pair $(\alpha_n(K,a),\beta_n(K,a))$ relative to (K,a), we have

$$\begin{aligned}&\sum_{n\geq 0}\alpha_n(K,a)\frac{(aq/d;q)_n}{(d;q)_n}(-d)^n q^{\binom{n}{2}}\sum_{r=n}^{\infty}\frac{(d,b,c;q)_r}{(Kd/a;q)_r}\frac{(Kq/bc)^r}{(q;q)_{r-n}(aq;q)_{r+n}}\\&=\frac{(Kq/b,Kq/c;q)_\infty}{(Kq/bc,K;q)_\infty}\sum_{n\geq 0}\beta_n(K,a)\frac{(aq/d,b,c;q)_n}{(Kq/b,Kq/c,Kd/a;q)_n}(1-Kq^{2n})\left(-\frac{Kd}{bcq}\right)^n q^{n(n+3)/2}.\end{aligned} \tag{5.9}$$

Proof. Substituting Rowell's conjugate Bailey pair into the definition (5.2) gives

$$\Gamma_n = \sum_{r=n}^{\infty} \frac{1}{(q;q)_{r-n}(aq;q)_{r+n}} \frac{(d,b,c;q)_r}{(Kd/a;q)_r} \left(\frac{Kq}{bc}\right)^r.$$

Multiplying this identity by

$$\frac{(Kq/bc,K;q)_\infty}{(Kq/b,Kq/c;q)_\infty} \frac{(aq/d;q)_n}{(d;q)_n} (-d)^n q^{\binom{n}{2}}$$

and using (5.6) gives

$$\gamma_n(K,a) = \sum_{r=n}^{\infty} \frac{(K/a;q)_{r-n}(K;q)_{r+n}}{(q;q)_{r-n}(aq;q)_{r+n}} \delta_r(K,a).$$

Then substituting this conjugate WP–Bailey pair into Lemma 5.1 and dividing by the infinite-product factor in (5.8) on both hand sides, it leads to (5.9). □

Specially, setting $K = aq$, (5.3) reduces to

$$\beta_n(aq,a) = \sum_{r=0}^{n} \alpha_r(aq,a).$$

For a given nonnegative integer ℓ, it is direct to see that the following is a trivial WP–Bailey pair relative to (aq,a)

$$\alpha_n(aq,a) = \begin{cases} 1, & n=\ell, \\ 0, & n\neq\ell, \end{cases} \qquad \beta_n(aq,a) = \begin{cases} 0, & 0\le n<\ell, \\ 1, & n\ge\ell. \end{cases}$$

Substituting it into (5.9) yields the following master identity.

Theorem 5.3. *For $\ell \ge 0$,*

$$\begin{aligned} &\frac{(aq/d;q)_\ell}{(d;q)_\ell}(-d)^\ell q^{\binom{\ell}{2}} \sum_{r=\ell}^{\infty} \frac{(d,b,c;q)_r}{(q;q)_{r-\ell}(aq;q)_{r+\ell}(dq;q)_r} \left(\frac{aq^2}{bc}\right)^r \\ &\quad = \frac{(aq^2/b,aq^2/c;q)_\infty}{(aq^2/bc,aq;q)_\infty} \sum_{r=\ell}^{\infty} \frac{(aq/d,b,c;q)_r}{(aq^2/b,aq^2/c,dq;q)_r}(1-aq^{2r+1}) \left(-\frac{ad}{bc}\right)^r q^{r(r+3)/2}. \end{aligned} \tag{5.10}$$

The first consequence of Theorem 5.3 is an extension of the truncated version of Euler's pentagonal number theorem (1.8) due to Ding and Sun [9].

Corollary 5.4. *For every positive integer ℓ,*

$$\begin{aligned} &\frac{1}{(aq;q)_\infty} \sum_{r=\ell}^{\infty} \frac{(aq;q)_r}{(q;q)_r}(1-aq^{2r+1})(-a)^r q^{r(3r+1)/2} \\ &\quad = (-1)^\ell a^\ell q^{(3\ell^2+\ell)/2}(aq;q)_\ell \sum_{n\ge 0} \frac{(q^\ell;q)_n a^n q^{n^2+(2\ell+1)n}}{(q;q)_n(q;q)_{n+\ell}(aq;q)_{n+2\ell}}. \end{aligned} \tag{5.11}$$

Proof. First take the limits $b, c \to \infty$ on both hand sides of (5.10) by applying

$$(b, c; q)_r \left(\frac{1}{bc}\right)^r \longrightarrow q^{r^2-r}.$$

Next letting $d \to 1$ and noting that for $r \geq \ell$,

$$\lim_{d\to 1} \frac{(aq/d; q)_\ell}{(d; q)_\ell} \frac{(d; q)_r}{(dq; q)_r} = (aq; q)_\ell \frac{(q^\ell; q)_{r-\ell}}{(q; q)_r},$$

then the left-hand side of (5.10) becomes the right-hand side of (5.11) after replacing $r = n + \ell$. Meanwhile, the right-hand side of (5.10) tends to the left-hand side of (5.11), which completes the proof. □

It is notable that when $a = 1$ in Corollary 5.4, it reduces to (1.8) directly.

Two further special cases of Theorem 5.3 produce companion forms of the classical truncated Gauss square-number and triangular-number series.

Corollary 5.5. *For every positive integer ℓ,*

$$\begin{aligned}
&\frac{(-q; q)_\infty}{(q; q)_\infty} \sum_{r=0}^{\ell-1} (-1)^r (1 - q^{2r+1}) q^{r^2} \\
&\quad = 1 + (-1)^{\ell-1} \frac{(-q; q)_\infty}{(q; q)_\infty} \sum_{n\geq 0} (-1)^n \frac{(1 + q^\ell)}{(1 + q^{n+\ell})} \frac{(q; q)_{n+\ell} q^{n(n+1)/2+\ell n+\ell^2}}{(q; q)_n (q; q)_{n+2\ell}},
\end{aligned} \tag{5.12}$$

and

$$\begin{aligned}
&\frac{(-q; q^2)_\infty}{(q^2; q^2)_\infty} \sum_{r=0}^{\ell-1} (-1)^r (1 - q^{2r+1}) q^{2r^2+r} \\
&\quad = 1 + (-1)^{\ell-1} \sum_{n\geq 0} \frac{(q^{2\ell}; q^2)_n (-q; q^2)_{n+\ell} q^{n^2+2(\ell+1)n+2\ell^2+\ell}}{(q^2; q^2)_n (q^{\ell+2}; q^2)_n (q^2; q^2)_{n+2\ell}}.
\end{aligned} \tag{5.13}$$

Proof. First taking $a = 1$, $b = q$, and $d = -1$ in Theorem 5.3, and then letting $c \to \infty$, we obtain the following truncated identity on square numbers

$$\sum_{r=\ell}^{\infty} (-1)^r (1 - q^{2r+1}) q^{r^2} = (-1)^\ell \frac{(-q; q)_\ell}{(-1; q)_\ell} \sum_{n\geq 0} \frac{(-1)^n (-1; q)_{n+\ell} (q; q)_{n+\ell} q^{n(n+1)/2+\ell n+\ell^2}}{(-q; q)_{n+\ell} (q; q)_n (q; q)_{n+2\ell}},$$

after substituting $r = n + \ell$. Since

$$\sum_{j=-\infty}^{\infty} (-1)^j q^{j^2} = \sum_{r=0}^{\infty} (-1)^r (1 - q^{2r+1}) q^{r^2},$$

multiplying the truncated identity by $(-q; q)_\infty/(q; q)_\infty$ and using (1.2) proves (5.12) by simplifying.

To prove the second identity, first substitute q by q^2 in Theorem 5.3. Setting $a=1$ and $b=-q$, and taking $c\to\infty$, $d\to 1$, it gives the truncated identity on triangular numbers

$$\frac{(-q;q^2)_\infty}{(q^2;q^2)_\infty}\sum_{r=\ell}^{\infty}(-1)^r(1-q^{2r+1})q^{2r^2+r}$$
$$=(-1)^\ell(q^2;q^2)_\ell\sum_{n\geq 0}\frac{(q^{2\ell};q^2)_n(-q;q^2)_{n+\ell}q^{n^2+2(\ell+1)n+2\ell^2+\ell}}{(q^2;q^2)_n(q^2;q^2)_{n+\ell}(q^2;q^2)_{n+2\ell}},$$

where the truncated sum on the left hand is derived by using

$$\frac{(-q;q^2)_r}{(-q^3;q^2)_r}(1-q^{4r+2})=(1+q)(1-q^{2r+1}),\quad (-q^3;q^2)_\infty=\frac{(-q;q^2)_\infty}{1+q}.$$

Note that by pairing the terms indexed by r and $-r-1$ in (1.3), its bilateral series can be expressed as $\sum_{r\geq 0}(-1)^r(1-q^{2r+1})q^{2r^2+r}$. Applying (1.3) and simplifying, it implies (5.13). □

Acknowledgments

This work is supported by the National Natural Science Foundation of China (No. 12571351, 12071235), Tianjin Natural Science Foundation (No. 24JCZDJC01390) and the Fundamental Research Funds for the Central Universities (No. 63261129).

References

[1] A.K. Agarwal, G.E. Andrews, and D.M. Bressoud, The Bailey lattice, *J. Indian Math. Soc. (N.S.)* **51** (1987), 57–73.
[2] G.E. Andrews, *The Theory of Partitions*, Cambridge Mathematical Library, Cambridge University Press, Cambridge, 1998.
[3] G.E. Andrews and A. Berkovich, The WP-Bailey tree and its implications, *J. London Math. Soc.* (2) **66** (2002), 529–549.
[4] G.E. Andrews and M. Merca, The truncated pentagonal number theorem, *J. Combin. Theory Ser. A* **119** (2012), 1639–1643.
[5] G.E. Andrews and M. Merca, Truncated theta series and a problem of Guo and Zeng, *J. Combin. Theory Ser. A* **154** (2018), 610–619.
[6] G.E. Andrews, A. Schilling, and S.O. Warnaar, An A_2 Bailey lemma and Rogers–Ramanujan-type identities, *J. Amer. Math. Soc.* **12** (1999), 677–702.
[7] W.N. Bailey, Identities of the Rogers–Ramanujan type, *Proc. London Math. Soc.* (2) **50** (1948), 1–10.
[8] W. Chu, Durfee rectangles and the Jacobi triple product identity, *Acta Math. Sinica (N.S.)* **9** (1993), 24–26.
[9] X. Ding and L.H. Sun, Truncated theta series from the Bailey lattice, *Adv. Appl. Math.* **167** (2025), 102884.
[10] J. Dousse, F. Jouhet, and I. Konan, Bilateral Bailey lattices and Andrews–Gordon type identities, *SIGMA* **21** (2025), 032, 32 pp.
[11] S. Fu and D. Tang, On certain unimodal sequences and strict partitions, *Discrete Math.* **343** (2020), 111650.
[12] W. Fulton, *Young Tableaux: With Applications to Representation Theory and Geometry*, London Mathematical Society Student Texts, Vol. 35, Cambridge University Press, Cambridge, 1997.

[13] G. Gasper and M. Rahman, *Basic Hypergeometric Series*, 2nd ed., Encyclopedia of Mathematics and its Applications, Vol. 96, Cambridge University Press, Cambridge, 2004.
[14] V.J.W. Guo and J. Zeng, Two truncated identities of Gauss, *J. Combin. Theory Ser. A* **120** (2013), 700–707.
[15] G. James and A. Kerber, *The Representation Theory of the Symmetric Group*, Encyclopedia of Mathematics and its Applications, Vol. 16, Addison–Wesley, Reading, MA, 1981.
[16] L.W. Kolitsch and M. Burnette, Interpreting the truncated pentagonal number theorem using partition pairs, *Electron. J. Combin.* **22** (2015), no. 2, Paper P2.55.
[17] R. Mao, Proofs of two conjectures on truncated series, *J. Combin. Theory Ser. A* **130** (2015), 15–25.
[18] M.J. Rowell, A new general conjugate Bailey pair, *Pacific J. Math.* **238** (2008), 367–385.
[19] D. Shanks, A short proof of an identity of Euler, *Proc. Amer. Math. Soc.* **2** (1951), 747–749.
[20] R.P. Stanley, *Enumerative Combinatorics, Vol. 1*, 2nd ed., Cambridge Studies in Advanced Mathematics, Vol. 49, Cambridge University Press, Cambridge, 2012.
[21] C. Wang and A.J. Yee, Truncated Jacobi triple product series, *J. Combin. Theory Ser. A* **166** (2019), 382–392.
[22] S.O. Warnaar, Supernomial coefficients, Bailey's lemma and Rogers–Ramanujan-type identities. A survey of results and open problems, *Sém. Lothar. Combin.* **42** (1999), Art. B42n, 22 pp.
[23] E.X.W. Xia, A.J. Yee, and X. Zhao, New truncated theorems for three classical theta function identities, *European J. Combin.* **101** (2022), 103470.
[24] A.J. Yee, A truncated Jacobi triple product theorem, *J. Combin. Theory Ser. A* **130** (2015), 1–14.